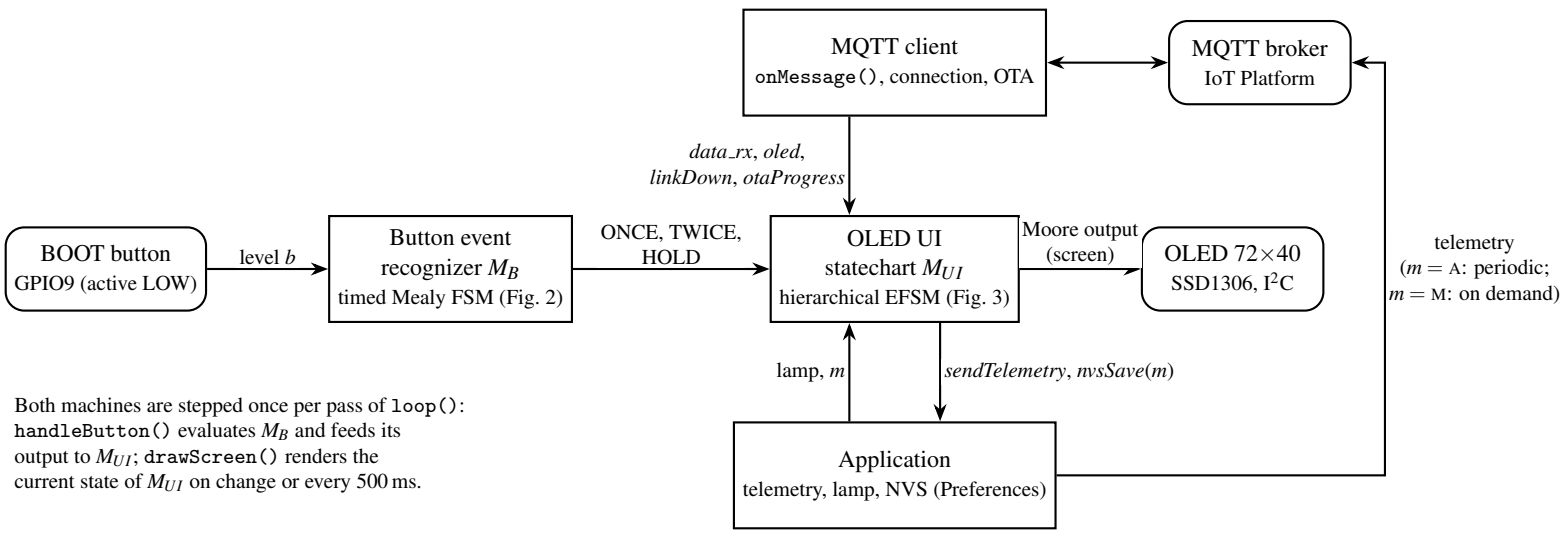
MQTT client
onMessage(), connection, OTA
MQTT broker
IoT Platform
data_rx, oled,
linkDown, otaProgress
BOOT button
GPIO9 (active LOW)
level b
Button event
recognizer M_B
timed Mealy FSM (Fig. 2)
ONCE, TWICE,
HOLD
OLED UI
statechart M_{UI}
hierarchical EFSM (Fig. 3)
Moore output
(screen)
OLED 72×40
SSD1306, I²C
telemetry
(m = A: periodic;
m = M: on demand)
lamp, m
sendTelemetry, nvsSave(m)
Application
telemetry, lamp, NVS (Preferences)
Both machines are stepped once per pass of loop():
handleButton() evaluates M_B and feeds its
output to M_{UI}; drawScreen() renders the
current state of M_{UI} on change or every 500 ms.